

VISION Resistivity				Schlumberger	
Dual Frequency					
Recorded Mode, Run 13					
Company:		Equinor			
Well:		31/5-7			
Field:		Eos			
Rig Name:		West Hercules			
Country:		Norway			
Latitude:		60° 34' 35.14" N		UWID:	
Longitude:		3° 26' 36.13" E		Rig Name:	
Block:		31/5		Rig Type:	
FL:		Norwegian North Sea			
FL1:		Section: 8 1/2 in.			
FL2:		Job no.: 19SCA0085			
Log Measured From: - Drill Floor: 31.00 m Permanent Datum: - Mean Sea Level Ground Level: 307.00 m					
Acquisition Dates:		13-Jan-2020 -- 15-Jan-2020		Other Services:	
Log Interval:		2699.00(m) -- 2916.00(m)		Direction & Inclination (Surveys)	
Index Types:		Measured Depth		Annular Pressure While Drilling	
Index Scales:		1:200		geoVISION Bit Resistivity	
Depth Source:		Driller's Depth			
Depth Sensor:		3rd Party Depth			
Print Type:		Final			
Spud Date:		02-Dec-2019			

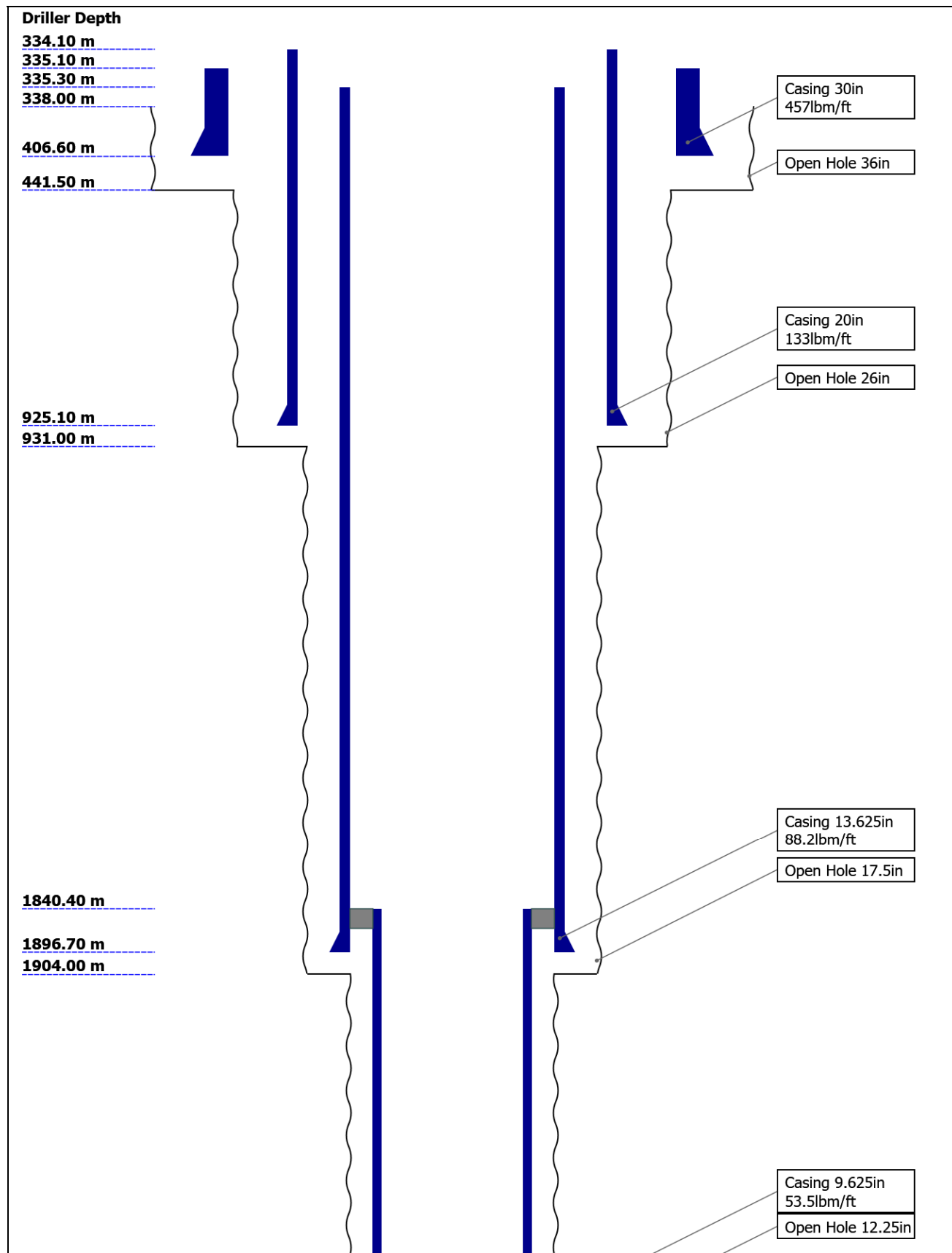
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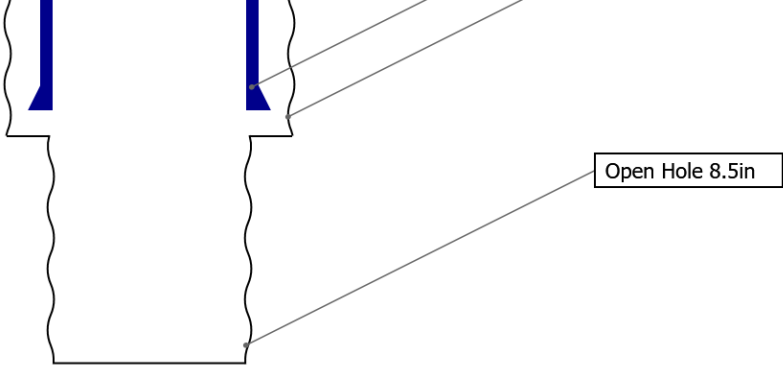
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Well Sketch



2583.00 m
2584.00 m

2915.00 m



Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2915	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	457	133	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

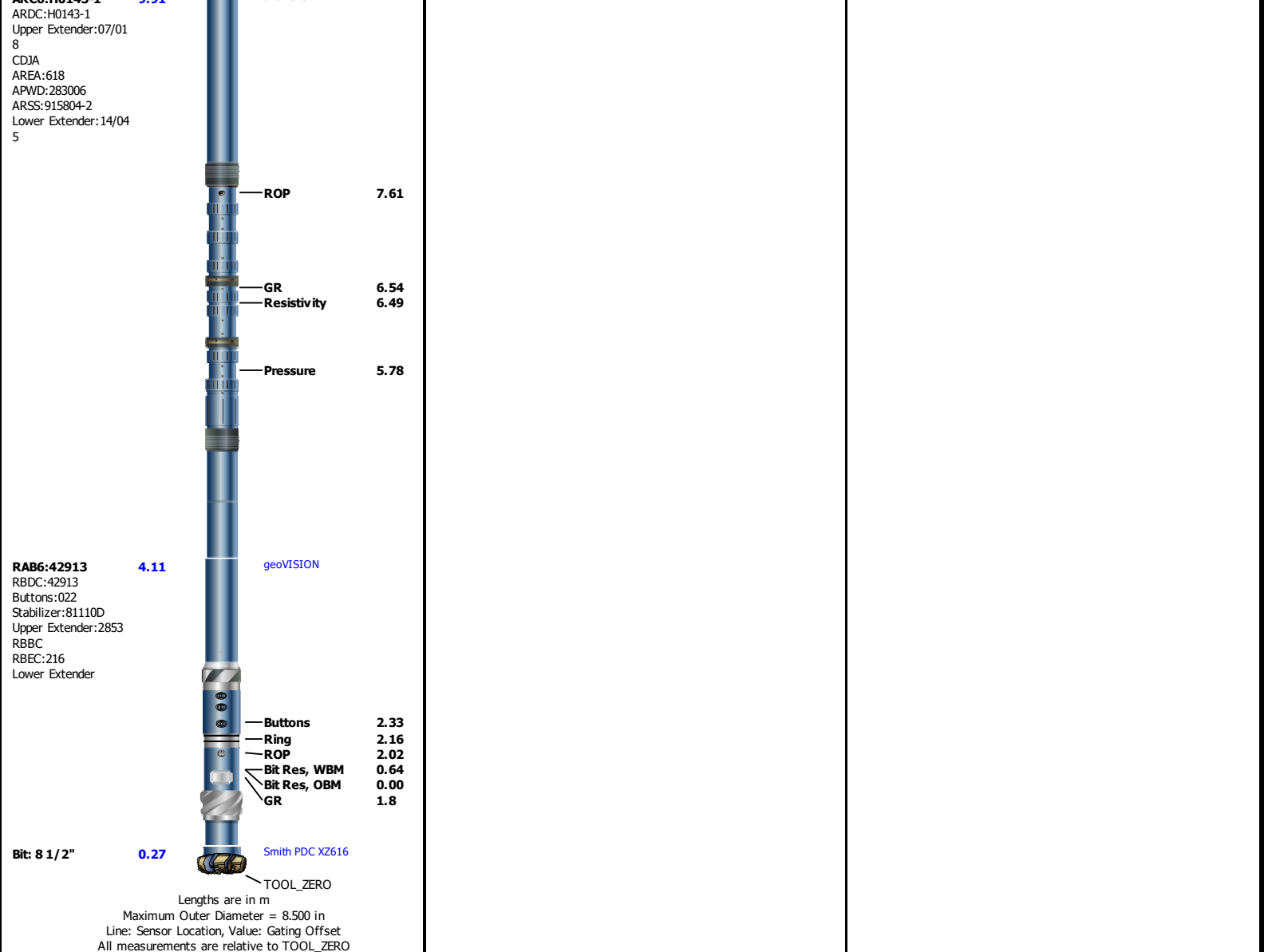
Operational Run Summary

Parameter (unit)	Run 13					
Date Log Started	13-Jan-2020					
Time Log Started	19:44:45					
Date Log Finished	15-Jan-2020					
Time Log Finished	10:55:25					
Bit Size (in)	8.500					
Bit Start Depth (m)	2784.00					
Bit Stop Depth (m)	2915.00					
Top Log Interval (m)	2699.00					
Bottom Log Interval (m)	2914.28					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	M.Bergenkull					
Witnessed By	R.Nordstokke					
Service Order Number	19SCA0085					

Borehole Fluids

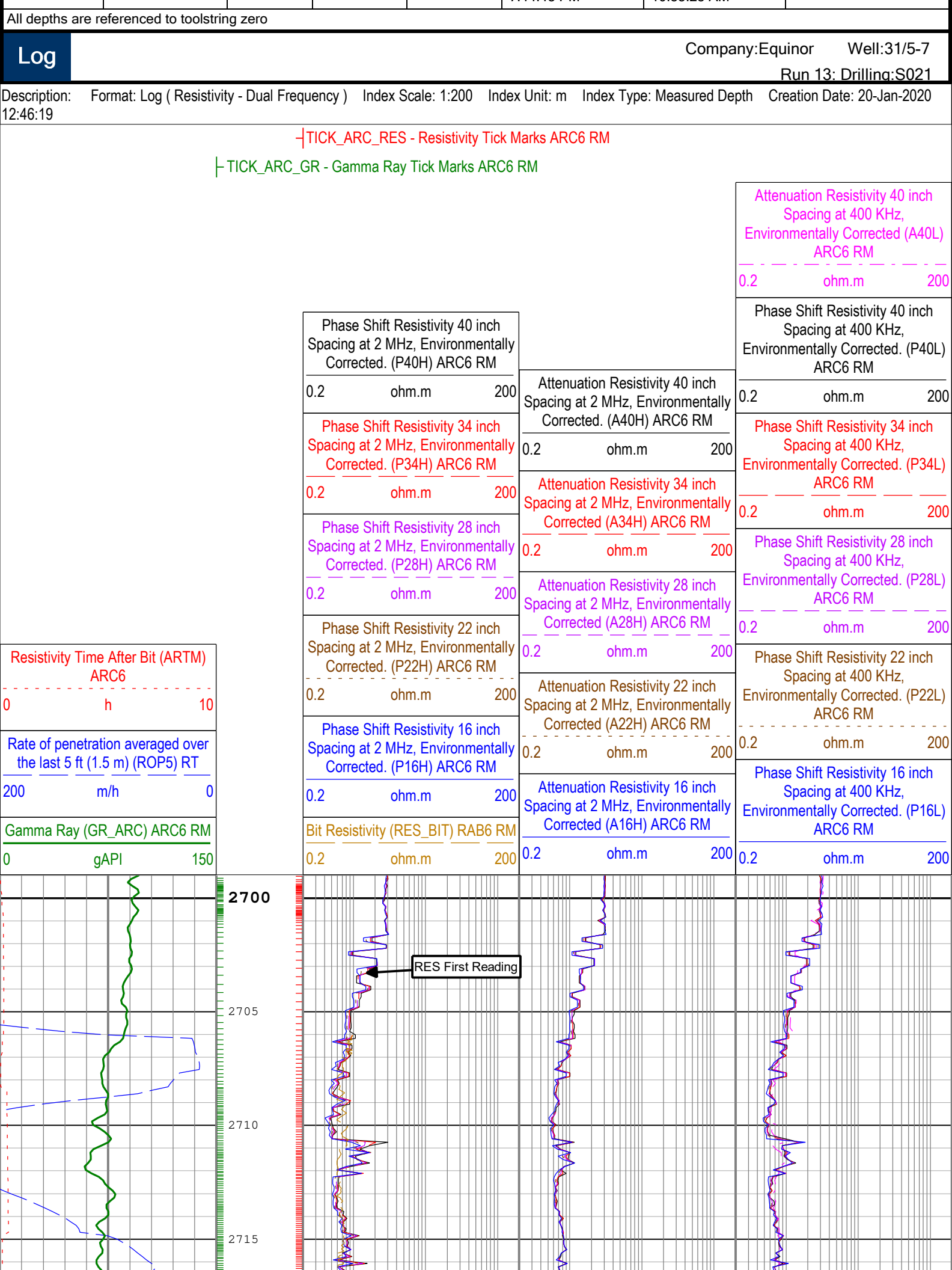
Parameter(unit)	Run 13					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	51					
Source of Sample	Active Tank					
Salinity (ppm)	79402					
Density (g/cm3)	1.15					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.09 @ 18.9					
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 17.9					
RMC @ Meas Temp (ohm.m@degC)	0.1 @ 20					
RM @ BHT (ohm.m@degC)	0.05 @ 51					
RMF @ BHT (ohm.m@degC)	0.04 @ 51					
RMC @ BHT (ohm.m@degC)	0.06 @ 51					
Total Solid (%)						
High Gravity Solids (%)						

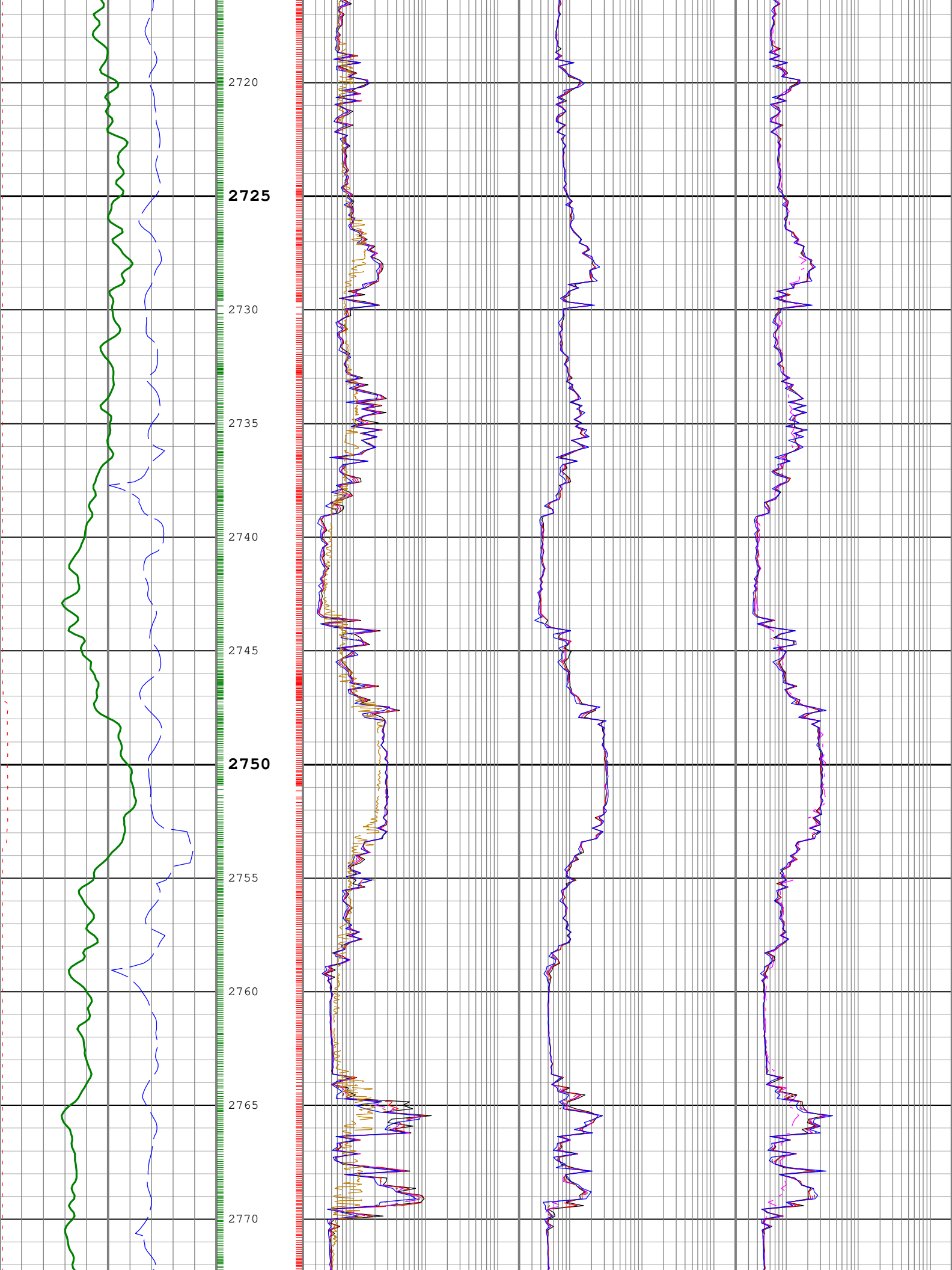
Run 13: Toolstring			Run 13: Remarks
Equip name TELE675:E4388 MSSU675:30848 Upper Extender MDC675:E4388 MMA:2507 MDI:3666 PMVC:236 PMEA:803 MTA:2162 MTK675:004 MSSD675:1005482-1 Lower Extender, Long	Length 18.37	MP name TeleScope	Offset
All depths referenced to driller's depth. See EOWR for depth tracking details.			
All data from tool memory. All data acquired while drilling.			
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.			
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.			
Data acquired while reaming down over cored interval from 2710 to 2784 m bit depth.			
Run 13, 8 1/2 in. section and well TD at 2915 m.			
ARC6:H0143-1	9.91	arcVISION	

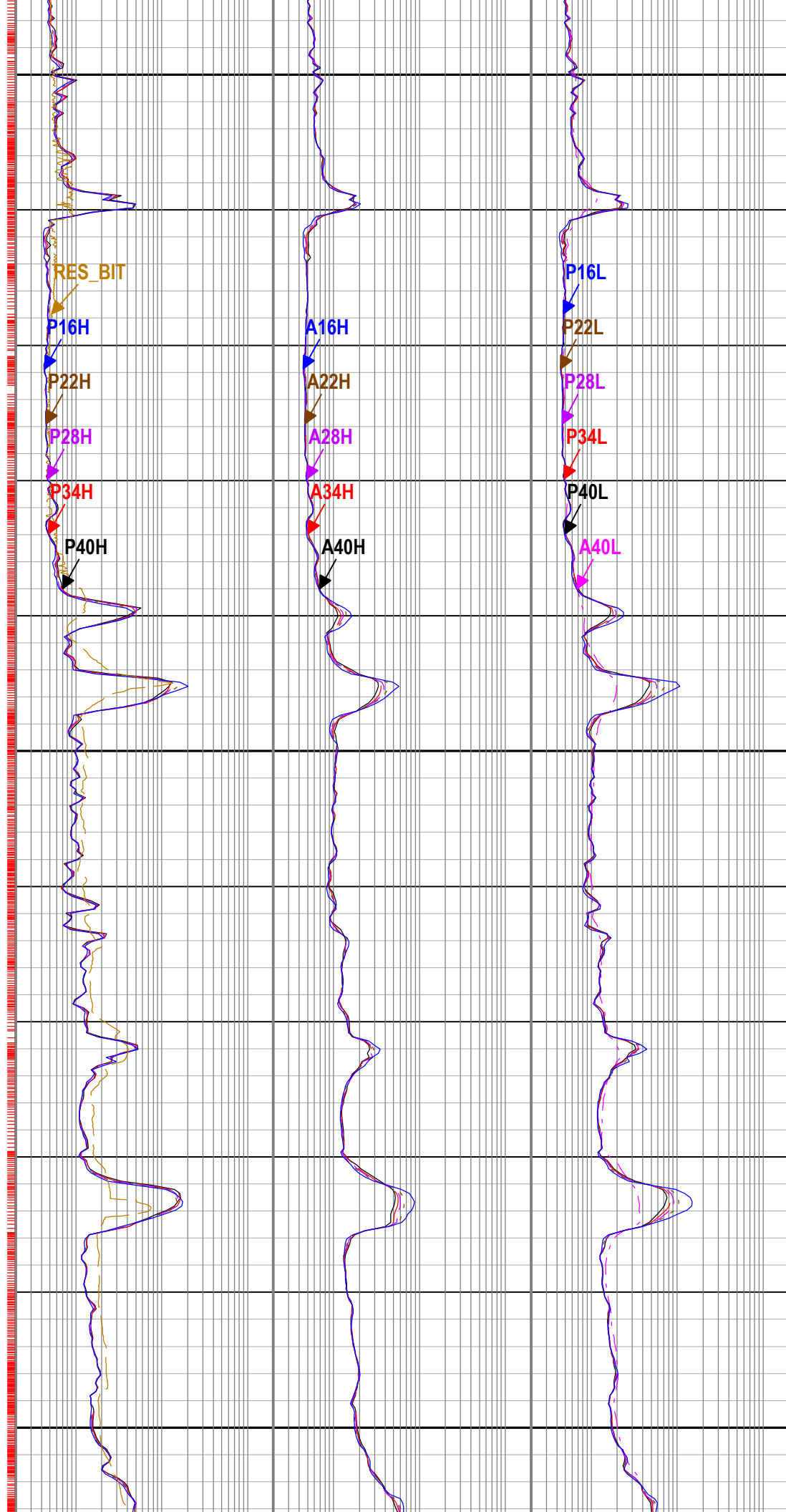
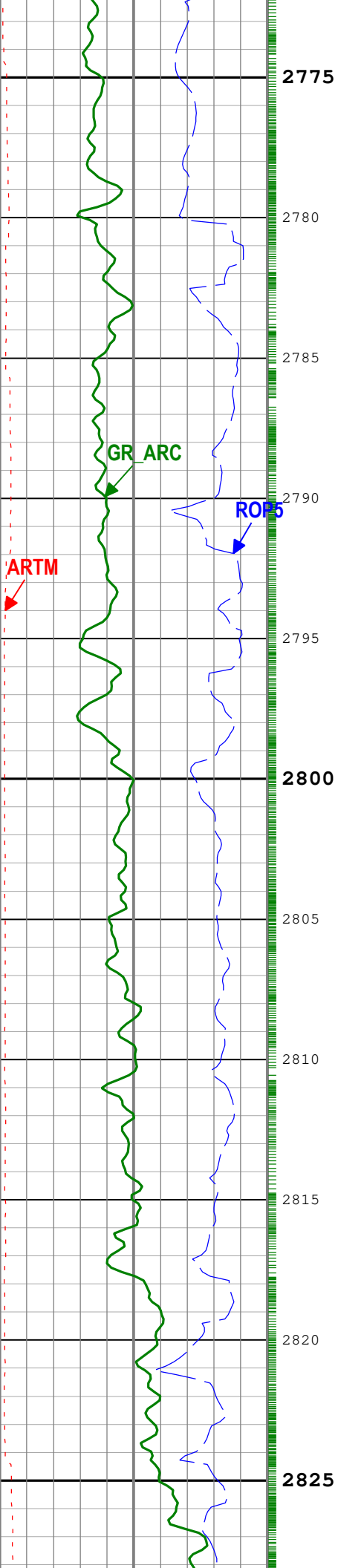


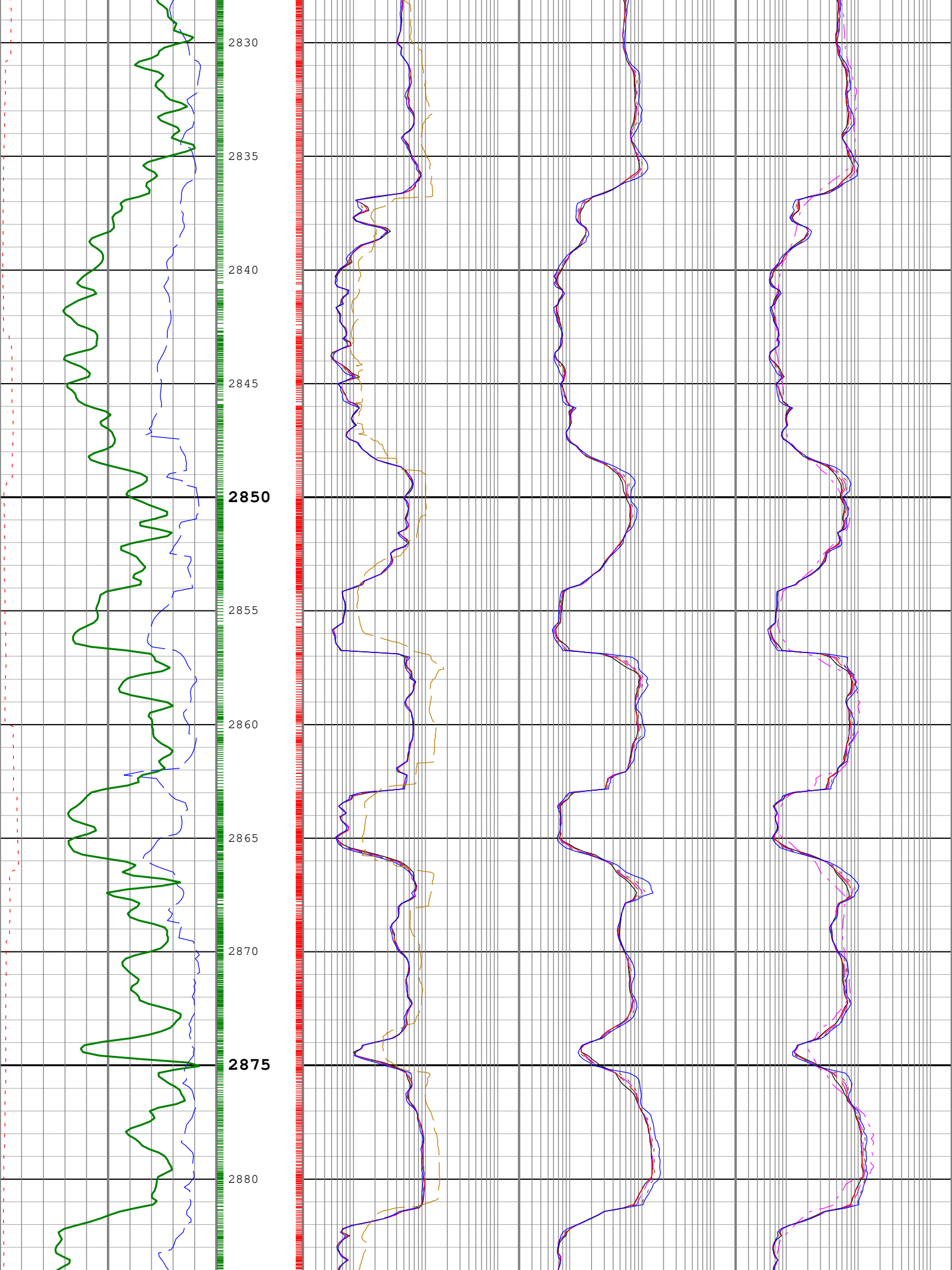
Run 13			
VISION Resistivity - Dual Frequency			
Software Version			
Acquisition System		Version	
Maxwell 2019.2		9.2.113335.3100	
Application Patch		DnM_TestKit-XceedCPU41-2019.2_9.2.114181	
Computation	Description		Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family		9.2.113335.3100
ARC6GammaRay	ARC6 Gamma Ray Computation Package for both Real-time and Recorded Mode		9.2.113335.3100
GVRRes	GVR Resistivity Computation Package for both Real-time and Recorded Mode		9.2.113335.3100
Tool Interface	System Version	Loaded Version	
HSPM	2019.0c.001	9.2.113335.3100	
Tool Elements	Description	Software Version	Firmware Version
RBEC	Electronics Chassis Assembly for RAB6-C	9.2.113335.3100	V8.8B
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 6.75 Inch Tool Drill Collar	9.2.113335.3100	V9.7B

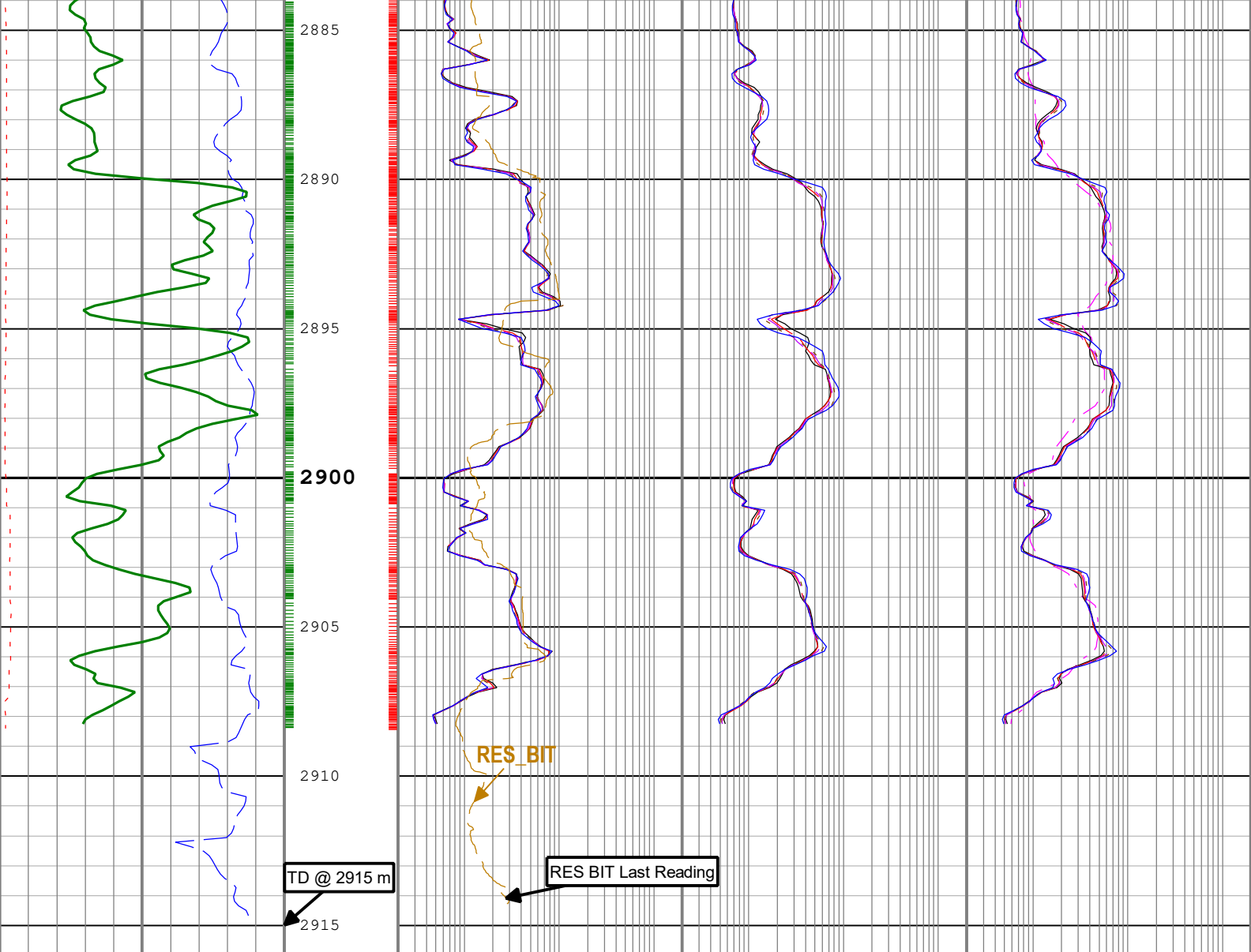
Pass Summary							
Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 13	Drilling	Down	181.74 m	2914.93 m	13-Jan-2020 7:44:45 PM	15-Jan-2020 10:55:25 AM	Yes











Resistivity Time After Bit (ARTM) ARC6 0h10	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC6 RM 0.2ohm.m200
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200m/h0	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC6 RM 0.2ohm.m200	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM 0.2ohm.m200
Gamma Ray (GR_ARC) ARC6 RM 0gAPI150	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC6 RM 0.2ohm.m200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM 0.2ohm.m200
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC6 RM 0.2ohm.m200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM 0.2ohm.m200
	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM 0.2ohm.m200	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC6 RM 0.2ohm.m200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM 0.2ohm.m200
	Bit Resistivity (RES_BIT) RAB6 RM 0.2ohm.m200		

Channel Processing Parameters

Run 13: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC6	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	6	%
BHT	Bottom Hole Temperature	Borehole	51	degC
BS	Bit Size	DNMSESSION	8.5	in
DFD	Drilling Fluid Density	Borehole	1.15	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	
GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC6	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC6	1	ohm.m
MST	Mud Sample Temperature	Borehole	18.9	degC
RMS	Resistivity of Mud Sample	Borehole	0.09	ohm.m
TEMP_SEL_RAB	RAB Temperature Selection	RAB6	Tool	

Tool Control Parameters

Calibration Report

RAB6 (GeoVision Resistivity 675) Calibration - Run 13

Primary Equipment :	Electronics Chassis	RBEC	216
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M21V - Resistivity

Master (Time Frame File):		16:32:26 08-Aug-2019						
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit		
Monitor 2 at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250		

M22V - Resistivity

Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

M01V - Resistivity

Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9950	1.0250	

M02V - Resistivity

Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T2 Calibration Coefficient		Master	1.0000	0.9750	0.9980	1.0250	

R1V - Resistivity

Master (Time Frame File):		16:32:26 08-Aug-2019						
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit		

Ring at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	
R2V - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	
BDM1 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0070	1.0250	
BDM2 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0100	1.0250	
BMM1 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9990	1.0250	
BMM2 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	
BSM1 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	
BSM2 - Resistivity							
Master (Time Frame File): 16:32:26 08-Aug-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	
PGR - Gamma Ray: Blanket							
Master (Time Frame File): 14:06:21 18-Dec-2019							
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.0000	0.7500	0.9838	1.2500	

ARC6 (Array Resistivity Compensated 675) Calibration - Run 13							
Primary Equipment :							
Elec. Chassis HP w/o AIM Receiver			AREA		618		
RESAIRCAL - Resistivity: Air							
Master (Time Frame File):		15:26:50 07-Dec-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	7.951	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.997	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	4.584	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.904	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.142	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.927	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.871	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.878	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.879	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.856	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	7.970	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.989	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	4.593	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.889	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.160	5.600	

Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	-0.666	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	0.584	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	-0.629	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	0.582	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	-0.643	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File):		10:23:30 06-Jun-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.580	1.100	1.250	

Company:	Equinor	
Well:	31/5-7	
Field:	Eos	
Rig Name:	West Hercules	
Country:	Norway	



VISION Resistivity

Dual Frequency

Recorded Mode, Run 13